

MORON AB, Spain

WMO: 083970

Lat: 37.164N

Lon: 5.611W

Elev: 87

StdP: 100.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.8	2.0	-3.9	2.7	6.4	-1.8	3.3	6.4	11.4	14.1	10.0	14.3	1.5	90	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.6	39.1	23.0	37.7	22.4	35.9	21.9	24.5	35.6	23.6	34.3	22.9	33.2	3.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	16.0	28.0	20.2	15.1	26.9	19.9	14.8	26.5	74.1	35.8	70.8	34.4	68.0	33.3	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.2	7.1	-1.6	42.3	2.3	2.1	-3.2	43.7	-4.5	45.0	-5.8	46.1	-7.4	47.6		
(5)			-2.4	26.1	2.2	1.1	-3.9	27.0	-5.2	27.6	-6.4	28.3	-7.9	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	10.4	11.5	14.2	16.3	19.9	24.0	26.9	27.5	24.0	20.4	14.5	11.5		
	DBStd	6.54	2.59	2.69	2.68	2.85	3.27	2.97	2.71	2.41	2.69	2.74	3.05	2.82		
	HDD10.0	62	25	14	3	0	0	0	0	0	0	0	3	17		
	HDD18.3	1003	245	191	128	72	22	1	0	0	0	12	119	213		
	CDD10.0	3153	38	57	134	191	306	419	524	542	420	322	138	62		
	CDD18.3	1052	0	0	2	12	69	170	266	283	170	75	4	0		
	CDH23.3	12624	0	1	68	232	913	2067	3419	3530	1709	652	31	1		
(13)	CDH26.7	6677	0	0	6	58	387	1086	2034	2073	817	213	3	0		
(14)	Wind	WSAvg	2.8	2.6	2.7	2.9	3.1	2.9	3.0	2.9	2.8	2.5	2.6	2.7		
(15) Precipitation	PrecAvg	632	81	65	56	70	42	9	1	2	35	85	84	102		
	PrecMax	1229	344	260	235	217	134	63	4	14	122	243	234	434		
	PrecMin	283	0	0	0	3	1	0	0	0	1	0	7	0		
	PrecStd	210	72	54	53	49	36	15	1	3	27	57	59	109		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	22.2	27.6	30.8	35.9	39.2	41.8	41.1	38.0	33.1	26.2	22.0		
		MCWB	13.8	14.4	16.9	18.5	21.7	22.6	23.9	24.0	22.6	19.7	17.1	14.6		
	2%	DB	18.1	20.9	25.2	28.3	33.0	36.8	39.2	39.1	35.2	31.0	24.0	19.2		
		MCWB	13.3	14.4	15.9	17.4	19.8	21.7	22.9	23.1	21.4	19.2	16.7	14.4		
	5%	DB	17.0	19.1	23.2	26.0	30.9	34.8	37.8	37.2	33.2	29.1	22.1	18.0		
		MCWB	13.3	13.5	15.1	16.6	18.9	21.0	22.5	22.5	21.0	18.8	16.3	14.2		
	10%	DB	15.9	17.2	21.1	23.9	28.8	32.9	35.9	35.8	31.4	27.1	20.2	16.9		
		MCWB	12.8	12.8	14.4	16.0	18.2	20.5	22.1	21.9	20.4	18.4	15.7	14.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	16.7	18.7	20.0	22.6	24.5	25.7	25.9	24.0	22.2	20.5	17.5		
		MCDB	17.8	19.8	24.5	27.2	32.5	35.2	37.8	37.8	33.5	27.0	22.1	18.7		
	2%	WB	15.1	15.5	17.1	18.6	21.0	23.1	24.4	24.6	22.8	21.0	18.6	16.3		
		MCDB	16.6	18.3	22.3	25.7	30.1	33.5	36.2	35.5	31.8	26.2	21.4	17.5		
	5%	WB	14.2	14.7	16.0	17.5	19.8	22.1	23.5	23.7	22.0	20.2	17.4	15.3		
		MCDB	16.0	17.4	21.1	24.1	28.2	32.1	35.0	34.2	30.1	25.5	20.5	17.0		
	10%	WB	13.1	13.7	15.1	16.5	18.8	21.2	22.6	23.0	21.2	19.5	16.4	14.2		
		MCDB	15.3	16.6	19.6	22.3	26.8	30.4	33.7	33.1	28.6	24.8	19.5	16.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.7	12.7	12.8	14.0	15.3	16.3	15.6	13.5	12.1	10.8	10.5		
		MCDBR	12.5	14.4	16.9	17.2	18.5	18.7	19.3	18.1	16.8	15.9	13.2	11.5		
		MCWBR	8.5	8.9	9.4	8.7	8.3	8.0	8.1	7.8	7.2	7.5	7.6	7.5		
	5% WB	MCDBR	9.1	11.2	13.7	14.9	16.0	16.7	17.3	16.3	14.1	12.1	10.1	8.3		
		MCWBR	7.0	7.7	8.2	8.0	7.9	7.7	7.9	7.6	6.7	6.3	6.6	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.344	0.380	0.380	0.395	0.407	0.391	0.405	0.396	0.373	0.339	0.324		
	taud		2.481	2.419	2.312	2.334	2.320	2.303	2.336	2.302	2.352	2.427	2.486	2.490		
	Ebn at Noon		869	887	881	900	888	874	885	865	854	843	838	837		
	Edh at Noon		82	98	120	124	128	130	125	127	114	97	81	77		
(44) All-Sky Solar Radiation	RadAvg		2.59	3.41	4.56	5.67	6.83	7.76	7.92	6.97	5.37	3.84	2.80	2.30		
	RadStd		0.30	0.44	0.48	0.42	0.50	0.31	0.17	0.21	0.32	0.34	0.26	0.24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+62

Nomenclature: See separate page